Monday Jan-21-2019

From: **To:**

,,

CERTIFICATE OF CONFORMANCE

Serial #	Heat # / Identifier	Customer PO #	Description	Class	End Connection	Shell Test	Seat Test
61-615204	P7402	1113812	4" 2345	900	FLG	3375	2475
61-615206	P7483	1113812	4" 2345	900	FLG	3375	2475
61-615156	P6427	1113812	4" 2345	900	FLG	3375	2475
61-615205	P7483	1113812	4" 2345	900	FLG	3375	2475

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, API Q1 9th edition, and API 6D 24th edition (if applicable to the API monogram). All valves/parts/sealant meet all the requirements of the customer purchase order.

Quality Engineer
Kenny Horne

SAMCO METALS AND ALLOYS PVT.LTD.

ISO 9001 - 2015. TUV CERT.REG.NO:44 100 094292 AD-WO TUVNORD APPROVAL
KANIYAMBADI,VELLORE Pin :632 102, Phone :0416-2230221, 2230222, 2917614, Fax :0416-2230459, email :samco@md1.vsnl.net.in
INDIA.

www.samcoalloys.com

CERTIFICATE OF MATERIAL AND TESTING FOR STEEL CASTING
(As per EN 10204-3.1)

Customer : FLOWSERVE US INC
1511 JEFFERSON STREET
SULPHUR SPRINGS TX 75482

Reference. 161705 Dt. 31/07/2018

T.C. No. : **P7402/38**
T.C.Date : **31/08/2018**
Melt No. : **P7402**
Date of Pouring : **17/08/2018**

Specification : **ASTM A216-2016 GR : WCC & RMC 90012 REV.13**

MELTING PROCESSED : BY BASIC INDUCTION

HEAT TREATMENT Details : Normalised, Raised to 940°C soaked for 3½ Hours and then cooled in still air.

SM :MANUFACTURER'S Symbol

CHEMICAL COMPOSITION - %

Element Requirment	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	Al	CE
Minimum	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.060	0.430
Actual	**0.210**	**0.990**	**0.470**	**0.018**	**0.008**	**0.070**	**0.030**	**0.010**	**0.010**	**0.006**	**0.040**	**0.395**

MECHANICAL PROPERTIES

Requirments	Yield Strength		Tensile Strength		%Elongation	% Reduction of Area	Hardness BHN (HBW)	Bend Test	Impact Test in (Joules) (10mm X 10mm X 55mm)			
	ksi	Kgf/sq.mm	ksi	Kgf/sq.mm	GL :50 mm			D = 3T				
	0.2% Proof Stress		-		-	-	-	at 90 °	CVN 2mm 45 °			
Minimum	45	31.56	70	49.47	22.00	35.00	---					
Maximum	---	---	95	66.81	---	---	237		1	2	3	Avg
Actual	53	36.92	81	56.92	34.60	59.79	163,163	QK	--	--	--	--

DETAILS OF CASTING POURED

SL.No.	Description	Drawing No.	Part No.	Quantity	
1	BODY 4"2345	D-487876 REV.S	487876WCC	8	

Remarks : We certify that the material was manufactured, sampled, tested and inspected in accordance with the material specification and the purchase order and was found to meet the requirements.
Castings have been MPI performed as per the RMC 90012 REV.13. Results found satisfactory.
Visual inspection carried out as per MSS SP 55 - 2011. Castings Degassed as per the RMP 61496 REV.A.
Castings are free from Radioactive contaminations / radiation.

We Certify the particular entered herein are correct and the material satisfies the specification .
ASTM A216-2016 GR : WCC & RMC 90012 REV.13

for SAMCO METALS AND ALLOYS PVT.LTD.

IRFAN AHMED K.M.M (Works Inspector)

CHANDRASEKARAN.D (Executive (QC))

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CERTIFICATE OF MATERIAL AND TESTING FOR STEEL CASTING
(As per EN 10204-3.1)

Customer : FLOWSERVE US INC
1511 JEFFERSON STREET
SULPHUR SPRINGS TX 75482

Reference. 161705 Dt. 31/07/2018

T.C. No. : **P7483/38**
T.C.Date : **31/08/2018**
Melt No. : **P7483**
Date of Pouring : **24/08/2018**

Specification : **ASTM A216-2016 GR : WCC & RMC 90012 REV.13**

MELTING PROCESSED : BY BASIC INDUCTION

HEAT TREATMENT Details : Normalised, Raised to 940°C soaked for 3½ Hours and then cooled in still air.

SM :MANUFACTURER'S Symbol

CHEMICAL COMPOSITION - %

Element Requirment	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	Al	CE
Minimum	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.060	0.430
Actual	**0.210**	**0.980**	**0.460**	**0.024**	**0.006**	**0.050**	**0.050**	**0.020**	**0.010**	**0.005**	**0.050**	**0.392**

MECHANICAL PROPERTIES

Requirments	Yield Strength		Tensile Strength		%Elongation	% Reduction of Area	Hardness BHN (HBW)	Bend Test	Impact Test in (Joules) (10mm X 10mm X 55mm)			
	ksi	Kgf/sq.mm	ksi	Kgf/sq.mm	GL :50 mm			D = 3T				
	0.2% Proof Stress		-		-	-	-	at 90 °	CVN 2mm 45 °			
Minimum	45	31.56	70	49.47	22.00	35.00	---					
Maximum	---	---	95	66.81	---	---	237		1	2	3	Avg
Actual	52	36.52	81	57.12	31.28	63.13	163,163	QK	--	--	--	--

DETAILS OF CASTING POURED

SL.No.	Description	Drawing No.	Part No.	Quantity	
1	BODY 6"4185½	D-78452 REV.13	00078452	2	
2	BODY 4"2345	D-487876 REV.S	487876WCC	6	

Remarks : NDE Details : RT No : SMA08451, SMA08452
We certify that the material was manufactured, sampled, tested and inspected in accordance with the material specification and the purchase order and was found to meet the requirements.
Castings have been MPI performed as per the RMC 90012 REV.13. Results found satisfactory.
Visual inspection carried out as per MSS SP 55 - 2011. Castings Degassed as per the RMP 61496 REV.A.
Castings are free from Radioactive contaminations / radiation.

We Certify the particular entered herein are correct and the material satisfies the specification .
ASTM A216-2016 GR : WCC & RMC 90012 REV.13

for SAMCO METALS AND ALLOYS PVT.LTD.

IRFAN AHMED K.M.M (Works Inspector)

CHANDRASEKARAN.D (Executive (QC))

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CERTIFICATE OF MATERIAL AND TESTING FOR STEEL CASTING
(As per EN 10204-3.1)

Customer : FLOWSERVE US INC
1511 JEFFERSON STREET
SULPHUR SPRINGS TX 75482

Reference. 161110 Dt. 21/05/2018

T.C. No. : **P6427/38**
T.C.Date : **30/06/2018**
Melt No. : **P6427**
Date of Pouring : **01/06/2018**

Specification : **ASTM A216-2016 GR : WCC & RMC90012 REV.13**

MELTING PROCESSED : BY BASIC INDUCTION

HEAT TREATMENT Details : Normalised, Raised to 940°C soaked for 3½ Hours and then cooled in still air.

SM :MANUFACTURER'S Symbol

CHEMICAL COMPOSITION - %

Element Requirment	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	Al	CE
Minimum	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.060	0.430
Actual	**0.210**	**1.030**	**0.470**	**0.023**	**0.014**	**0.090**	**0.030**	**0.040**	**0.010**	**0.010**	**0.050**	**0.412**

MECHANICAL PROPERTIES

Requirments	Yield Strength		Tensile Strength		%Elongation	% Reduction of Area	Hardness BHN (HBW)	Bend Test	Impact Test in (Joules) (10mm X 10mm X 55mm)			
	ksi	Kgf/sq.mm	ksi	Kgf/sq.mm	GL :50 mm			D = 3T				
	0.2% Proof Stress		-		-	-	-	at 90 °	CVN 2mm 45 °			
Minimum	45	31.56	70	49.47	22.00	35.00	---					
Maximum	---	---	95	66.81	---	---	237		1	2	3	Avg
Actual	49	34.48	78	54.77	30.86	56.38	156,156	QK	--	--	--	--

DETAILS OF CASTING POURED

SL.No.	Description	Drawing No.	Part No.	Quantity	
1	BODY 4"2345	D-487876 REV.S	487876WCC	8	

Remarks : We certify that the material was manufactured, sampled, tested and inspected in accordance with the material specification and the purchase order and was found to meet the requirements.
Castings have been MPI performed as per the RMC 90012 REV.13. Results found satisfactory.
Visual inspection carried out as per MSS SP 55 - 2011.
Castings are free from Radioactive contaminations / radiation.

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ASTM A216-2016 GR : WCC & RMC90012 REV.13

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Reference. 161705 Dt. 31/07/2018

T.C. No. : **P7483/38**
T.C.Date : **31/08/2018**
Melt No. : **P7483**
Date of Pouring : **24/08/2018**

Specification : **ASTM A216-2016 GR : WCC & RMC 90012 REV.13**

MELTING PROCESSED : BY BASIC INDUCTION

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CHEMICAL COMPOSITION - %

Element Requirment	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	Al	CE
Minimum	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.200	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.060	0.430
Actual	**0.210**	**0.980**	**0.460**	**0.024**	**0.006**	**0.050**	**0.050**	**0.020**	**0.010**	**0.005**	**0.050**	**0.392**

MECHANICAL PROPERTIES

Requirments	Yield Strength		Tensile Strength		%Elongation	% Reduction of Area	Hardness BHN (HBW)	Bend Test	Impact Test in (Joules) (10mm X 10mm X 55mm)			
	ksi	Kgf/sq.mm	ksi	Kgf/sq.mm	GL :50 mm			D = 3T				
	0.2% Proof Stress		-		-	-	-	at 90 °	CVN 2mm 45 °			
Minimum	45	31.56	70	49.47	22.00	35.00	---					
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SL.No.	Description	Drawing No.	Part No.	Quantity	
1	BODY 6"4185½	D-78452 REV.13	00078452	2	
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